Enhanced thermoelectric figure of merit (ZT) of Te-doped $FeSb_2$ nanocomposite

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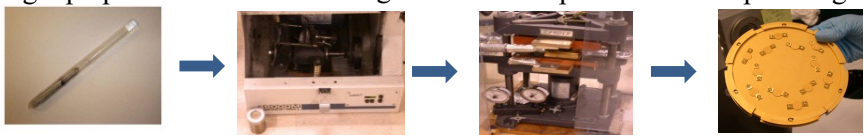
Introduction:

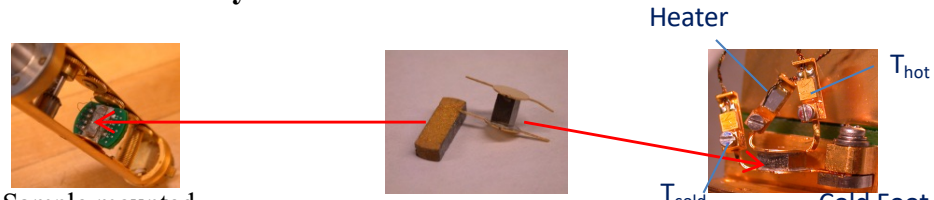
Doping Te on the Sb side of $FeSb_2$ is shown to improve thermoelectric efficiency for single crystal samples [1]. In our work we apply a nanostructuring approach to an optimized stoichiometry, $FeSb_{1.84}Te_{0.16}$. The combination of doping and nanostructuring result in a reduced thermal conductivity and increased power factor. The peak value of ZT for the optimized sample was 0.022 at 100 K, an increase of 62 % over its single crystal counterpart.

Sample Preparation:

Ingot preparation → Ball milling → DC hot pess → Sputtering



Measurement system:



Sample mounted on horizontal rotator of PPMS

Samples cut in bar shape

Sample mounted on TTO puck of PPMS

Results and Discussion:

- Decreased thermal conductivity of nanostructured samples is further reduced due to point-defect scattering.
- The peak value of the Seebeck coefficient in doped samples decreases as an indication of increased carrier density.
- A semiconductor to metal transition is induced by doping and leads to a metallic ground state.
- Significant power factor enhancement occurs in Te doped samples when compared to undoped nanostructured samples.
- The over all effect was to enhance ZT by 62 %.

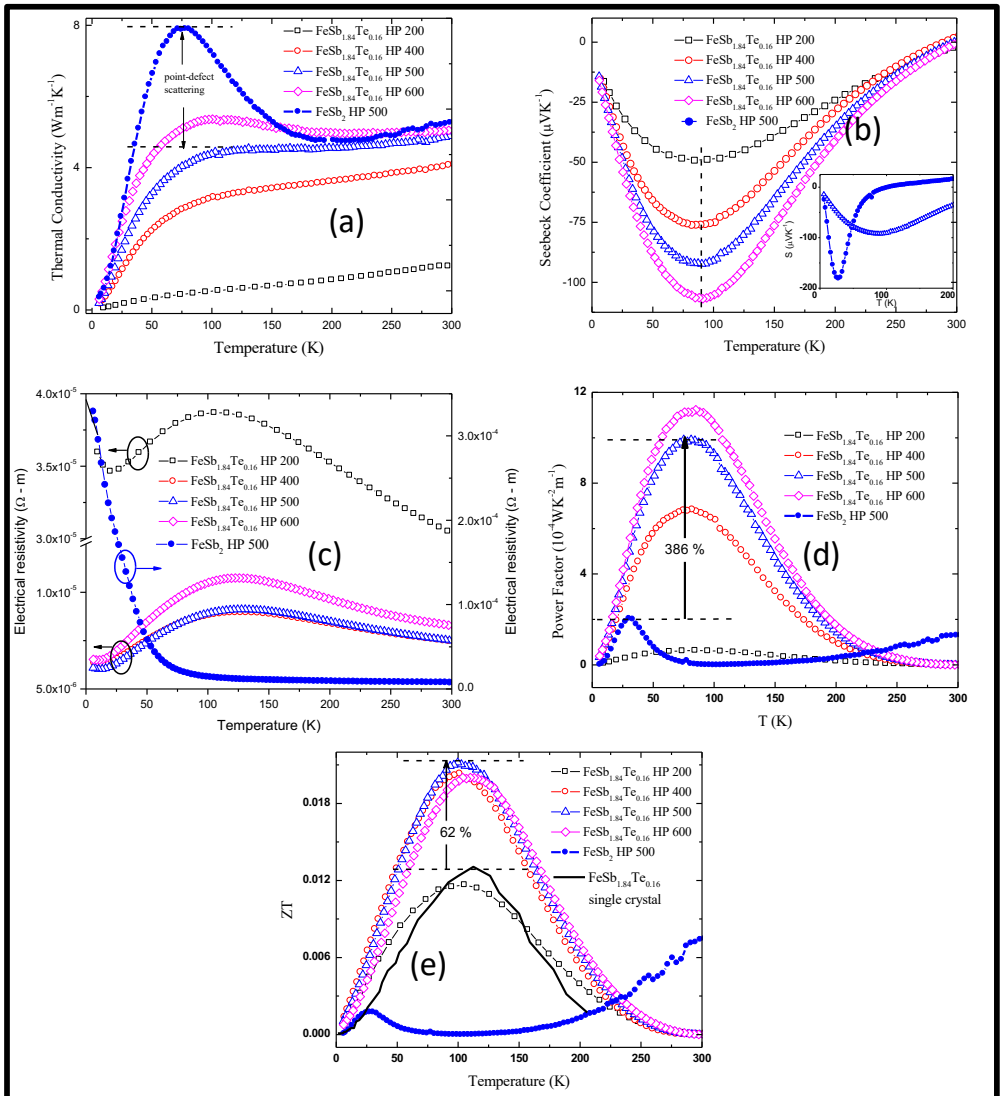


Figure 1: The thermal conductivity (a), Seebeck coefficient (b), electrical resistivity (c), power factor (d) and ZT (e) as a function of temperature.

Conclusion:

ZT of the nanostructured $FeSb_{1.84}Te_{0.16}$ sample increases by 62 % over its single crystal counterpart. Our results indicate that optimized doping combined with nanostructuring could be a promising way to improve ZT in phonon dominated system like $FeSb_2$.

Reference [1] Sun *et al.*; *Appl. Phys. Lett.* **98**, 2011.